MILITARY SPECIFICATION

TRUCK, MAINTENANCE: WITH VERTICAL LIFT PLATFORM, GASOLINE ENGINE DRIVEN, 11,000 POUNDS GVW, 4x2, COMMERCIAL

This specification is approved for use by the US Army Tank Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline engine driven, four-wheel, two rear-wheel drive, 11,000 pounds gross vehicle weight (GVW) commercial trucks equipped with a maintenance body and a vertically telescoping four-section mast and fold-down platform. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Storage.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS51335	- Pintle Assembly, Towing - 18,000 Lbs. Capacity, Manual Release.
MS51317	- Light, Warning, Vehicular, Rotating, D.C.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards
1926.556 - Aerial Lifts.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of solicition.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI A92.2 - American National Standards for Vehicle-Mounted Elevating and Rotating Aerial Devices.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards.
ASTM Designation: B117 - Standard Method of Salt Spray (Fog) Testing (DoD adopted).

(Application for copies ASTM pubications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.
SAE Standards and Recommended Practices.

J245	- Engine Rating Code - Spark Ignition.
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).

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J588 - Turn Signal Lamps.
J589 - Turn Signal Switch.
J704 - Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
J1292 - Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1349 - Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All chassis items shall be as represented and rated in the chassis manufacturer's technical data and special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color shall be gloss yellow, matching color chip No. 13538 of FED-STD-595. Identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

3.1.1.1.1 Caution, instruction, operation and schematic plates. Caution, instruction, operation and schematic plates shall be furnished. Instruction plates shall indicate complete operating and lubricating instructions for the vehicle. Schematic plates shall be furnished and shall depict all valves, valve controls and related equipment. Plates shall conform to type III, composition A or composition C of MIL-P-514. Plates shall be fastened on the inside of the curbside vertical cabinet door.

3.1.1.1.2 Caution plates. Caution plates shall be furnished and mounted at a conspicuous location near the dual controls and in the driver's compartment. At least the following equipment operating information shall appear on the plates:

WARNING

(a) PLATFORM AND OUTRIGGERS MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED PLATFORM LOAD, HYDRAULIC OUTRIGGERS DOWN, NOT IN EXCESS OF _____ POUNDS.
(c) OVERALL TRAVEL HEIGHT WHEN PLATFORM IS IN STOWED POSITION _____.
(d) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 30 MPH.
(e) MAXIMUM LONGITUDINAL INCLINE _____ DEGREES.
(f) MAXIMUM TRANSVERSE SLOPE _____ DEGREES.
(g) USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID.

3.1.1.1.3 Warning signs. Warning signs or labels, as recommended by Manufacturers of Aerial Devices and Digger Derrick Council, shall be furnished and attached.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. All drain plugs installed in manual transmission and rear axle for draining lubricants shall be of the permanent magnet type.

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3.1.1.4 Heavy duty cooling system. When specified (see 6.2), a heavy duty cooling system shall be furnished. The system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.5 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.6 Towing devices. Not less than two hooks, loops, or pins, mounted to the frame or to the rigid members which are attached to the frame, for towing the vehicle, shall be furnished on the front of the vehicle.

3.1.1.7 Trailer towing package. When specified (see 6.2), a trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51335-1 and shall be installed on the chassis frame with reinforcements to transfer a tongue load of not less than 1,800 pounds and a drawbar load of not less than 18,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward but not more than four inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The safety chain attachment devices shall be capable of accommodating a standard grab hook (2-3/16 inches wide, 47/64 inches thick, 31/64-inch throat width) for a 3/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.8 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

3.1.1.9 Silicone brake fluid. When specified (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A tag shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

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3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body; vertical platform assembly in stowed position; a full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds) and a payload distributed throughout the length of the vehicle body, to provide not less than the specified GVW. The GVW shall be not less than 11,000 pounds.

3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle and component ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of the ratings must be submitted to the engineering office of the procuring activity. The ratings shall be established by the manufacturer of the equipment being verified.

3.2.8 Cab-to-axle. The cab-to-axle dimension shall be not less than 60 inches. The cab-to-axle dimension shall be measured from the back of the cab to the center of the rear axle.

3.2.9 Safety standards. The vehicle shall comply with the requirements of OSHA 1926.556, and with ANSI A92.2.

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3.2.10 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing, and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. The vehicle shall maintain a high speed of not less than 58 miles per hour (mph) on smooth, approximately level, hard surfaced roads. The vehicle shall be capable of ascending grades of at least 20 percent in low-speed range. Speed and gradeability requirements shall be met with the vehicle loaded to specified GVW.

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52, under all conditions of loading, up to specified GVW.

3.3.3 Parking brakes. A hand- or foot-actuated parking brake shall be furnished in addition to the service brakes. When a foot actuated parking brake is furnished, a hand lever shall be furnished to release the parking brake. The vehicle parking brake shall be capable of holding the vehicle loaded to specified GVW on a 30 percent slope.

3.3.4 Upsetting stability. With one and one-half times the specified load (see 3.6.2) placed on the work platform with the vehicle otherwise unloaded, and with the two outriggers extended, the telescopic mast shall be capable of placing and holding the loaded platform in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one-half times the specified load when any one of the vehicle wheels or outriggers lifts off of the ground. The stability test shall not cause any permanent deformation beyond the elastic limit of any component of the vehicle. The vehicle shall also be deemed unstable if, with twice the specified load (see 3.6.2), the inner tire of either rear wheel lifts off of the ground.

3.3.4.1 Stability on slopes. With one and one-third times the specified load (see 3.6.2) and the vehicle otherwise unloaded and with the vehicle not being leveled, on a 5-degree slope, facing upward and then downward, and then at right angles to the 5-degree slope, and with the outriggers extended, the telescopic mast shall be capable of placing and holding the loaded platform in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one-third times the specified load when any one of the vehicle's tires or outriggers lifts off of the ground.

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3.3.5 Wind resistance. With the specified load (see 3.6.2) in the platform and the vehicle otherwise unloaded and in elevated position, the vehicle shall be capable of withstanding a wind load of 7 pounds per square foot without upsetting, overturning, or permanent deformation of any component. The maximum area exposure for wind load shall be calculated at the body, with the mast and platform in the most critical position that would induce tipping. The resisting moment shall be greater than the wind load when the load is applied on the surface of the body, mast and platform.

3.3.6 Height. The boom shall be capable of raising the work platform with rated load (see 3.6.2), measured from ground level to the platform floor, to a height of not less than 29 feet.

3.3.7 Position stability (work platform). When the vehicle is on a 5 percent grade, without being leveled, the elevating mechanism shall hold the loaded work platform (see 3.6.2) in position within 2 inches per hour vertical settling. The settling shall be measured with hydraulic fluid stabilized at ambient air temperature.

3.4 Chassis components.

3.4.1 Engine. The engine shall be a liquid cooled, internal combustion, four-stroke cycle, gasoline type, with not less than six cylinders. The engine shall develop not less than 115 net brake horsepower at manufacturer's recommended governed speed and shall have a net torque capacity of not less than 215 pound feet. Engine net torque and net horsepower rating shall be determined in accordance with SAE J245 or SAE J1349. The engine shall develop the specified torque and horsepower when operated on unleaded fuel with a research octane rating of 91. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.1 Power plant heaters. When specified (see 6.2), coolant and battery heaters shall be provided. Heaters shall operate on 120 volts alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to

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Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed

(b) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperatures of the electrolyte to not more than 80°F.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Ignition system. An ignition system of 12-volt direct current (dc) potential shall be furnished. Unless otherwise specified (see 6.2), an alternator of not less than 35 amperes dc rated capacity which provides not less than 7 amperes dc output at normal engine idle speed shall be furnished.

3.4.2.2 Battery. The battery shall be of 12-volt potential. The reserve capacity rating, measured in accordance with SAE J537, shall be not less than 80 minutes. The cold cranking rating at 0°F, measured in accordance with SAE J537, shall be not less than 350 amperes. The battery shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

3.4.2.3 Lighting. All vehicle lights and reflectors shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d).

3.4.2.3.1 Turn signals. Turn signals shall be furnished. Turn signal lights may be combined with stoplights at rear and with parking lights at front. Taillights and stoplights may be incorporated into the same housing. Control for turn signals shall be self-cancelling type, with visible and audible type flash indicator. Turn signal lamps shall conform to SAE J588 and turn signal operating unit shall conform to SAE J589, class A.

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3.4.2.4 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67 (except compliance with 393.67(c), subparagraphs (1), (4)(ii), (11), 393.67(d) and 393.67(f) is not required). The capacity of the fuel tank shall be not less than 19 gallons.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil bath type air cleaner, oil capacity shall be not less than 1 pint.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. A spark arrester with an efficiency rating of not less than 80 percent when tested in accordance with SAE J350 shall be furnished, except on turbocharged engines.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. The transmission shall have not less than four forward speeds and one reverse speed. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with a power takeoff opening conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle and engine furnished, with clutch torque capacity exceeding maximum delivered engine torque.

3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff openings. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

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3.4.5.4 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type of vehicle furnished. The vehicle shall have a frame with reinforcements extending at least from the rear of the front-suspension, rear hanger bracket to the rear spring, front hanger bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased bending moment to at least equal the loads imposed with the vehicle body loaded to provide specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a total rated capacity per axle at least equal to the load imposed with the vehicle loaded to specified GVW. The vehicle shall be equipped with hydraulic double-acting shock absorbers at the front wheels. Suspension system ratings shall be adequate for the loads imposed during all static operating conditions.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. Gear ratios shall provide performance specified in 3.3.1. A single reduction rear axle shall be furnished.

3.4.9.1 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel(s) having traction when the opposite wheel(s) loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations, for the type and size of tires furnished. Tire, rim size, disc wheel size, and load range (ply rating) shall be the same for all wheels on each vehicle. All wheels, including the spare wheel, shall be interchangeable between the front and rear axles without the use of an adapter.

3.4.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed with the vehicle loaded to specified GVW. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

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3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16.1.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim identical to those furnished on the vehicle shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range as the tires furnished on the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.4.11 Brakes. Manufacturer's standard power brakes shall be furnished. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42, 393.45 through 393.48 and 393.52.

3.4.12 Cab. The manufacturer's standard full width cab shall be furnished. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with an external, key-operated lock. Drip rails shall be provided above the cab doors. Unless otherwise specified, the cab shall have upholstered, full width, adjustable seat and back or individual, adjustable driver's seat and individual passenger's seat. When specified (see 6.2), the cab shall be equipped with an individual, adjustable driver's seat and an individual passenger's seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior dome lighting shall be provided. Manufacturer's standard seat belt fastenings and two pairs of seat belts shall be installed.

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3.4.13 Steering. Power steering shall be provided.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. The windshield wipers shall be multi-speed type and operated by electric motor(s).

3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided. The rear bumper shall conform to 3.5.1. Rear end protection shall conform to Federal Motor Carrier Safety Regulation 393.86.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains and emergency reflective triangles shall be furnished. Stowage space for these tools may be furnished in the cab or in the cabinets. When stowage space for these tools is located outside of the cab or cabinets, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

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(a) Key-operated starting switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Ash receptacle
(h) Dual sunvisors
(i) Front door mounted armrest on driver's side
(j) Driver's compartment ventilator other than window.

3.4.20 Rearview mirrors. Outside rearview mirrors shall be provided on both sides. The outside rearview mirrors shall be of the Junior West Coast or low mount type incorporating a wide angle viewing system. The upper mirror shall have a flat reflective area of not less than 35 square inches. The lower mirror shall have a convex reflective area of not less than 20 square inches. The mirror assembly heads shall be adjustable and mounted to minimize obstruction of forward lines of vision.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished.

3.5 Body. The vehicle shall be equipped with a line maintenance and construction type body having minimum exterior dimensions specified in table I. The base of the mast shall be located in the loadspace behind the cab.

TABLE I. Body dimensions (minimum).

Body dimensions	Inches
Rear-of-cab to rear-axle dimensions	60
Overall length of body	108
Overall width of body	88
Loadspace width (exclusive of wheelwells)	43
Cabinet space (each side including wheelwells):	
Length	96
Height at front	42
Height at rear	38

3.5.1 Bumper and platform extension. A combination bumper and body platform extension shall be provided. Safety tread steel plate shall be provided on top of the platform extension.

3.5.1.1 Stirrup steps. Stirrup steps shall be furnished on each side of the combination bumper and platform extension. Grab handles shall be installed to assist personnel entering the body. Stirrup step and grab handles shall be furnished on the rear of both rear cabinets for access to the top of the cabinets.

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3.5.1.2 Cabinets. Cabinet headers shall be not less than 18 gage (0.0478 inch) high tensile steel, or 16 gage (0.0598 inch) carbon steel. Cabinet backs, partitions, and shelves shall be of not less than 20 gage (0.0359 inch) high tensile steel, or 18 gage (0.0478 inch) carbon steel. Cabinet doors shall be of not less than 20 gage (0.0359 inch) high tensile steel, or 18 gage (0.0478 inch) carbon steel, single thickness reinforced, or 22 gage (0.0299 inch) carbon steel, double panel reinforced. Doors shall be weatherproof and constructed to maintain fit and alinement in service. Doors shall have continuous rod, or hem type, heavy duty hinges, with noncorrosive hinge pins or sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of or plated with corrosion-resistant material. Horizontal doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 25 pounds on any square foot of area. Cabinet arrangement shall be manufacturer's standard and shall provide the following:

(a) Full height vertical cabinets equipped with bolted or screwed-on hooks
(b) Horizontal cabinets equipped with small parts bins with adjustable spacers, shelves and partitions, and a rubber-goods compartment
(c) Forward cabinets equipped with not less than two adjustable tray type shelves.

3.5.1.3 Maintenance body lighting. In addition to the lighting specified in 3.4.2.3, the following lighting and wiring shall be furnished:

(a) Lights with toggle switches and guards in each cabinet compartment;
(b) Lights on universal mounts and of sufficient candlepower to illuminate body loadspace and all equipment contained therein and all controls;
(c) Two 4-1/2 inch minimum diameter sealed beam floodlights with universal mounts shall be installed, one on each side of the top of the cab, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area;
(d) An amber, rotating, vehicular warning light in accordance with MS51317-4 shall be provided. The rotating light shall be mounted with reinforcement at the point of attachment on or near the left front corner of the cab for 360 degree visibility. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch;

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(e) When specified (see 6.2), a strobe type warning light shall be furnished in lieu of the rotating light specified in (d) above. The warning light shall be an integral solid-state unit, having a peak xenon tube output rating of not less than 1,000,000 candlepower and a flash rate of 90 ± 15 flashes per minute. The strobe light unit shall include a xenon flash tube with optic lens mounted in a weatherproof housing with protective clear or colored dome and corrosion-resistant metal base. Housing shall be not less than 8-1/2 inches high by 8-3/4 inches base diameter. Warning light color, mounting installation, wiring, switch and indicator light shall be as specified in (d) above;

(f) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292. All wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body framework.

3.5.1.4 Body mounting. The body shall be mounted in accordance with the manufacturer's recommendations. Breaker strips, when used, shall be tapered toward the front of the vehicle.

3.6 Vertical lift assembly. The lift assembly shall include a four-section telescoping mast and a work platform. The lift assembly shall be mounted in the bed of the truck directly behind the cab. The lift assembly shall be hydraulically actuated.

3.6.1 Lift construction and operation. The hydraulic lift shall be composed of four telescoping square steel sections. The lift shall be actuated by a hydraulic cylinder. The lift shall raise and lower at a rate of not more than one foot per second. The lowering speed shall be controlled by a hydraulic control valve and flow regulator. A manual adjustment to the holding valve shall be provided to lower the lift in case of power source failure.

3.6.2 Guardrail type work platform. The non-insulated, guardrail type work platform shall be of the fixed type with drop-down side extensions. The load capacity of the platform shall be not less than 700 pounds. The platform shall have two guardrails and a kickrail around the perimeter of the platform. The kickrail shall be not less than 5 inches in height; the lower guardrail shall be not less than 20 inches from the floor of the platform; and the top guardrail shall be not less than 42 inches from the floor of the platform. With the

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drop-down extensions of the platform locked in the stowed (up) position, a working space of not less than 36 inches by 70 inches shall be provided. The sides of the platform shall drop-down and shall extend to provide a working area of not less than 36 inches by 134 inches. A swinging gate, with a positive safety latch to hold the gateway closed, shall be provided to permit access to the platform. Protruding edges shall be padded with a resilient replaceable bumper.

3.6.2.1 Flooring. The flooring of the work platform shall be of the open grating type. No point in the flooring shall deflect more than 1/4 inch or be permanently deformed when subjected to a vertical load of 300 pounds covering an area of 25 square inches.

3.6.3 Clearance warning lights. Two amber strobe lights conforming to 3.5.1.3(e) shall be mounted to the underside of the platform. The lights shall be clearly visible to oncoming traffic. The warning lights shall be automatically actuated when the platform is raised.

3.6.4 Stowage of lift assembly. The stowed height of the lift assembly shall be not more than 7 feet, measured from the ground to the top of the mast. For stowage and travel, the work platform shall lower to rest on four rubber bumper pads mounted to the top of the cabinets. The drop-down extensions of the platform shall be folded up to the travel position.

3.6.5 Controls. A complete set of hand-operated controls for maneuvering the lift assembly shall be provided both in the platform and in the body. The controls shall be spaced at least four inches apart so as to permit operation with arctic gloves. A single lever control may be furnished in the platform. The placement of the controls shall prevent inadvertent actuation by the operator. Controls shall be located in a position easily accessible to the operator. Smooth operation, with no lurching of the lift assembly, shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. All lift and outrigger controls shall be the dead-man type operable with one hand. The controls in the body shall be easily accessible from the floor of the body and shall be provided for the operator. The body controls shall be capable of overriding the controls on the platform. A safety interlock system shall be provided to preclude operation of the lift when the outriggers are not extended. Ground controls shall be placed to minimize possibility of use by unauthorized personnel. A means shall be provided to lower and stow the boom, to retract the outriggers and to place the vehicle in travel position in case of power source failure. The switches, instruments and pressure gages on the lower control station

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shall be panel mounted, identified as to their function and installed in a manner to facilitate removal and service. Gages and instruments shall be shockproof, waterproof and weatherproof. Identification markings shall be on a photoetched, corrosion-resistant plate.

3.6.6 Outriggers. Two hydraulically operated outriggers shall be provided. Outriggers shall insure adequate stability with relation to ground while operating the platform under maximum loads. Outrigger cylinder hydraulic fittings, hoses and pilot check valves shall be protected by metal guards and shall be accessible for service. Outriggers shall be designed so that they may be extended over curbings and sidewalks when operating in narrow areas. Individual controls for each outrigger shall be furnished to permit leveling of the vehicle on inclined surfaces. Controls for the left outrigger shall be provided at the rear left corner of the vehicle and controls for the right outrigger shall be provided at the rear right corner of the vehicle. Automatic pilot check valve shall be provided with an automatic, positive spring-actuated lock, or an automatic pilot check valve shall be provided. The check valve shall hold the outrigger in the retracted position without settling, and in the event of loss of hydraulic system pressure, may be manually released.

3.6.6.1 Outrigger pads. Each outrigger shall be provided with a ground pad. Ground pads shall be of the hinged or fixed type.

3.6.6.2 Outrigger warning light. A red warning light shall be installed in the cab in a conspicuous location to warn the driver that the outriggers are extended.

3.6.6.3 Circular level. Two circular levels (Geir and Blume or equal), with 0 to 5 degrees reading shall be furnished. Circular levels shall be mounted in a protected and visible position at the outrigger controls. Circular levels shall indicate the degree of body level when the outriggers are extended. Circular levels shall be resistant to inclement weather and shall be corrosion-resistant. A warning plate (see 3.1.1.1.1 and 3.1.1.1.2) shall be mounted adjacent to each level to indicate the maximum longitudinal incline and the transverse slope, in degrees, on which the vehicle is capable of operating.

3.6.7 Hydraulic system. The hydraulic system shall consist of the components required to perform all hydraulic operations of the lift assembly and outriggers with the degree of control specified herein. Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. An automatic throttle actuator shall be furnished which increases the speed of the truck engine when the pump is being loaded. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service.

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3.6.7.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the lift and the outriggers shall be provided with pilot check or holding valves to prevent descent of the lift and retraction of the outriggers in case of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.6.7.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the lift assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fittings shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. The system shall be designed to preclude hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and for completely draining all hydraulic lines. A manual type bleeder shall also be provided as on-vehicle equipment on closed-center systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable. A pressure gage shall be provided on the discharge side of the hydraulic pump.

3.6.7.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The hydraulic pump shall be driven by a heavy duty power takeoff unit attached to the truck chassis transmission. Chain drives shall not be used. The pumping system shall include an adjustable pressure relief valve or unloading valve with a range of adjustment which allows raising or lowering of the mast and platform with any working or test load specified herein.

3.6.7.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be completely baffled. The

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reservoir shall be provided with a filler opening and removable oil strainer and a vented filler cap, accessibly located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to preclude fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain to prevent loss. Provision shall be made for the inspection and cleaning of the interior and for draining the reservoir. The reservoir drain shall be so located that the fluid will not drain on other components of the vehicle and shall allow recovery of the fluid in a container. The reservoir shall have a photoetched, corrosion-resistant plate to indicate type of hydraulic fluid furnished and the maximum fluid level in the reservoir with the platform in the travel position and the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to observe the level of the fluid. The tank shall be clean. A plate (see 3.1.1.1.2) shall be secured to the hydraulic fluid reservoir stating "USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID."

3.6.7.4.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

3.6.7.5 Oil filter and strainer. A full flow type, 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction line strainer in the hydraulic reservoir shall be provided. Filter and suction strainer shall be easily accessible for periodic cleaning or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow the maximum rated pump discharge.

3.6.7.6 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to preclude seal damage or approach to the flash point of the fluid during all operating conditions of the vehicle and use of the lift assembly for long periods of time. Excessive heating (more than 55°F above ambient in an ambient of 125°F, or equivalent in lower ambients) of the fluid shall not be acceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

3.6.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully-enclosed, and properly guarded.

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3.6.8.1 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and stowed in the cabinets.

3.7 Dissimilar metals. All dissimilar metals throughout the vehicle shall be insulated from one another to preclude galvanic action or electrolytic action.

3.8 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. Vehicles shall be lubricated, prior to delivery, with applicable grade of lubricant for the ambient air temperature encountered at the delivery point.

3.9 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete servicing of chassis, engine, running gear and mounted equipment with grades of lubricants or hydraulic fluids recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.10 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts and castings shall be commensurate with sound manufacturing standards for similar equipment. All metal and plastic materials used shall be free from delamination defects, and shall be prime material. Improper manufacturing or assembling practices and defective components or parts and assemblies which have been modified to overcome deficiencies shall not be furnished.

3.10.1 Welding. Adequate preparation of metal prior to welding of the body lift and platform components shall be provided. Surface of parts to be welded shall be free of rust, scale, paint, grease and other foreign matter. Welding rods shall be compatible with the metal being welded. All critical arc welds shall be magnetic particle or dye penetrant inspected. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding
(c) Undersize fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding

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(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack or intermittent welds for strength.

Weld penetration shall provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.10.2 and 3.10.3.

3.10.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body and lift components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.10.3 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.10.4 Factors. The factors listed below which are applicable shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspections.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to start of fabrication of the subfloor, body, base and lift assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

4.3.2 In-process inspection of body and components. The first complete assembled body and components (subframe, body, base, outriggers, lift assemblies and work platform) shall receive an in-process inspection by the Government prior to prime painting or application of protective coatings. The inspection shall be conducted to evaluate workmanship and to determine conformance of structural members to contractual requirements.

4.3.2.1 Structural inspection. The vehicle shall be tested in accordance with 4.4.8 and dye penetrant or magnetic particle inspection shall be performed on the critical areas of the welds prior to painting.

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4.4 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government to ascertain that the vehicle is adequate for the intended loading. Distribution of weight shall be computed by using the curb weight, the operator weight at 175 pounds, and a payload considered to be evenly distributed throughout the length of the vehicle body, to provide the specified GVW.

4.4.2 Road test. The first production vehicle shall be examined and road tested by the contractor to assure that the vehicle will operate in accordance with contractual requirements. The vehicle, loaded with payload and all the equipment specified herein, shall be driven at speeds up to at least 50 mph, for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 10 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all compartment and cab doors shall be opened and shut and there shall be no evidence of mud or dust inside; and there shall be no evidence of racking of the body.

4.4.3 Body treatment and painting. The certification regarding body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.4.4 Upsetting and range of movement test. With the truck resting on approximately level ground, without payload, and not fastened to any artificial base, the platform shall be loaded with twice the specified payload (see 3.6.2). The platform shall be elevated to its full height as indicated in 3.3.3. The inner dual tire shall be observed for conformance to the stability requirements of 3.3.3 with twice the load (see 3.6.2). The test shall be repeated with one and one-half times the specified payload (see 3.6.2) and the outriggers and all tires observed for conformance to the stability requirements of 3.3.3 with one and one-half times the load.

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Apply a within b°F of the oven temperature, but not less than 1 hour.

4.4.4.1 Stability on slopes test. The tests specified in 4.4.4 shall be repeated, except the truck shall be resting on a 5 degree slope, not leveled, in positions specified in 3.3.3.1.

4.4.5 Wind test. With the outriggers extended (see 3.6.6) and the rated capacity placed on the platform, the mast shall be extended to maximum height. The specified wind load, generated by artificial means or by an equivalent simulated load, shall be applied to the vehicle from either side (see 3.3.5).

4.4.6 Speed test. With the vehicle resting on approximately level ground, the platform shall be loaded with specified load (see 3.6.2). The platform shall be elevated to determine conformance to the requirements in 3.3.5 and 3.3.6.

4.4.7 Settling test. With the vehicle longitudinal on a 5 percent grade, without leveling the vehicle, and with specified load (see 3.6.2) on the platform, the platform shall be elevated. The position shall be held for one hour for maximum settling of 2 inches per hour, to determine conformance to requirements of 3.3.7.

4.4.8 Static conditions. With the mast at its most elevated position, and with three times the specified load (see 3.6.2) placed in or suspended from the platform, soluble penetrant and magnetic particle inspection areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle. Dye penetrant and magnaflux inspection shall be conducted prior to the first production vehicle inspection. Records of the tests and dye penetrant and magnaflux inspection witnessed by the Government shall be presented at the time of the first production vehicle inspection.

4.4.8.1 Critical areas. At least the following critical areas shall be inspected by dye penetrant or magnetic particle:

(a) Outriggers, and attaching areas
(b) Base, trunnion and yoke at attachment of mast and struts
(c) Structural steel of mast
(d) Yoke at attachment to platform and side brackets.

4.4.9 Performance test. The platform shall be subjected, but not limited, to the following tests:

(a) Check to determine whether lift assembly can be elevated without outriggers;

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(b) Determine if outriggers have locked automatically when in an extended position and in a retracted position;
(c) Operate equipment using each set of controls separately;
(d) Check to insure that the lower controls override the platform controls;
(e) Check to insure that the entire system can be operated smoothly and without jerk motions in any position;
(f) Check the hydraulic system including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions;
(g) Check hydraulic fluid temperature after system reaches steady state condition.

4.4.10 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

4.4.11 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample. The production sample shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4.12 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrected.

4.5 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.9 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for replacement of lamps, tree trimming, painting, window washing, and many other types of overhead construction and maintenance work.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Concealed markings, if required (see 3.1.1.1).
(d) Rustproofing, if required (see 3.1.1.2).
(e) Tropical rustproofing, if required (see 3.1.1.2).
(f) Heavy duty cooling system, if required (see 3.1.1.4).
(g) A trailer towing package, if required (see 3.1.1.7).
(h) Silicone brake fluid, if required and available (see 3.1.1.9).
(i) Power plant heaters, if required (see 3.4.1.1).
(j) Alternator capacity, if other than as specified (see 3.4.2.1).
(k) Automatic transmission, if required (see 3.4.5.3).
(l) Traction control, if required (see 3.4.9.1).
(m) A spare tire carrier, if required (see 3.4.10.3).
(n) A spare wheel or rim, if required (see 3.4.10.4).
(o) A spare tire assembly, if required (see 3.4.10.5).
(p) Tools, if required (see 3.4.16.1).
(q) Individual driver's and passenger's seats, if required (see 3.5.1.1).
(r) A strobe type warning light, if required (see 3.5.1.3(e)).

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Preparing activity:

Army - AT

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